

Assignment 1 – GANs

Deep Neural Networks for Computer Graphics

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May 25, 2026

1 Task 1.2a – Padding Derivation

We use a 2D convolution with kernel size $K = 4$ and stride $S = 2$ to downsample the spatial dimension by a factor of 2, i.e. we require $H_{out} = H_{in}/2$. The standard output-size formula is:

$$H_{out} = \left\lfloor \frac{H_{in} + 2P - K}{S} \right\rfloor + 1.$$

Substituting $K = 4$, $S = 2$ and setting $H_{out} = H_{in}/2$,

$$\frac{H_{in}}{2} = \frac{H_{in} + 2P - 4}{2} + 1 \iff H_{in} - 2 = H_{in} + 2P - 4 \iff \boxed{P = 1}.$$

2 Task 1.7 – DCGAN W&B Report

[Link to the DCGAN W&B Report](#)

3 Task 2.6 – CycleGAN W&B Report

[Link to the CycleGAN W&B Report](#)

4 Task 2.7 – Bonus: Identity Loss

We added an identity-loss term to the CycleGAN objective, exposed via the new flags `--use_identity_loss` and `--lambda_identity` (default $5.0 = 0.5 \lambda_{\text{cycle}}$):

$$\mathcal{L}_{\text{id}} = \|G_{X \rightarrow Y}(y) - y\|_1 + \|G_{Y \rightarrow X}(x) - x\|_1.$$

We expected it to help based on the proposal in the original CycleGAN paper (Zhu et al., 2017, sec. 5.2). The adversarial + cycle objective constrains the round trip but says nothing about what a generator should do when its input is already in the target domain. Without this term, the generators can introduce unnecessary color/style shifts; the identity loss explicitly discourages that.

W&B report. [Link to the Bonus W&B Report](#)